

Self Assessment 1

$$1. a) \frac{x^2 y^{-2}}{x^{-8}} = x^2 x^8 y^{-2} = \cancel{x^{10} y^{-2}} = x^{10} y^{-2} = \frac{x^{10}}{y^2}$$

$$b) \left(\frac{a^3}{2b^3} \right)^3 = \boxed{\frac{a^9}{8b^9}}$$

$$2. a) 2 \quad b) -4 \quad c) \frac{1}{2} \quad d) \pi$$

$$3. \begin{array}{r} 8x^2 + x + 1 \\ - (7x^2 - 8x - 15) \\ \hline \end{array} \rightarrow \begin{array}{r} 8x^2 + x + 1 \\ - 7x^2 + 8x + 15 \\ \hline x^2 + 9x + 16 \end{array}$$

$$4. \frac{8x-1}{x+4} - 2 = \frac{8x-1}{x+4} - \frac{2(x+4)}{x+4} = \frac{8x-1-2x-8}{x+4} = \frac{6x-9}{x+4}$$

$$5. x^2 + 2x - 3 = 0$$

$$(x+3)(x-1) = 0$$

$$x = -3$$

$$x = 1$$

$$6. 805n$$

$$7. |3x+4| < 5 \rightarrow -5 < 3x+4 \quad \text{and} \quad 3x+4 < 5$$

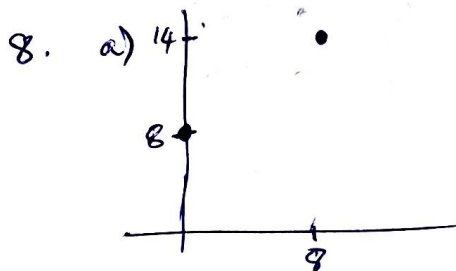
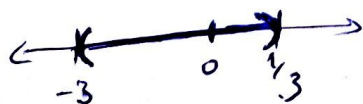
$$-5 < 3x+4 < 5$$

$$\frac{-9}{3} < \frac{3x}{3}$$

$$\underline{\underline{-3 < x}}$$

$$\frac{3x}{3} < \frac{1}{3}$$

$$\underline{\underline{x < \frac{1}{3}}}$$



$$b) \text{ distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad c) (4, 11)$$

$$= \sqrt{(8-0)^2 + (14-8)^2}$$

$$= \sqrt{64 + 6^2}$$

$$= \sqrt{64 + 36} = \sqrt{100} = \underline{\underline{10}}$$

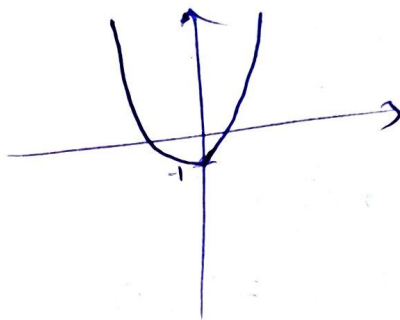
9. a) $\frac{y_2 - y_1}{x_2 - x_1} = \text{rate of change} = \frac{8 - 0}{0 - 2} = \frac{8}{-2} = \underline{\underline{-4}}$

b) $f(x) = ax + b$
 $a = -4$
 $b = 2$
 $f(x) = -4x + 2$

10. $f(x) = \sqrt{x-5}$ domain: $[5, \infty)$
range: $[0, \infty)$

11.

x	$f(x) = x^2 - 1$
± 5	24
± 4	15
± 3	8
± 2	3
± 1	0
0	-1



12. $h(t) = -t + \frac{5}{4}$ $t = -3$ $t = 1$

a) $h(-3) = -(-3) + \frac{5}{4}$
 $= 3 + \frac{5}{4}$
 $= \frac{17}{4}$

$h(1) = -(1) + \frac{5}{4}$
 $= -\frac{4}{4} + \frac{5}{4}$
 $= \frac{1}{4}$

b) average rate of change

$\frac{4}{-3-1} = \frac{4}{-4} = \underline{\underline{-1}}$

net change = $\frac{17}{4} - \frac{1}{4} = \frac{16}{4} = \underline{\underline{4}}$

13. $f(x) = (x+2)^2$ + 2 shifts graph left by 2 units

14. $f(x) = 4x - 2$

$f(f(x)) = 4(4x - 2) - 2$
 $= 16x - 8 - 2$

$f \circ f(x) = 16x - 10$

$f \circ f(-1) = -16 - 10 = \underline{\underline{-26}}$

$g(x) = 5 - x^2$

$g(g(x)) = 5 - (5 - x^2)^2$

$= 5 - (25 - 10x^2 + x^4)$

$g \circ g(x) = x^4 - 10x^2 - 20$

$g \circ g(-1) = 1 - 10 - 20$

$= \underline{\underline{-29}}$

15. $f^{-1}(f(6)) = 6$

16. $y = 2x^3 + 11x$ has no local maximums/minimums

17.

$$\begin{array}{r}
 x^3 + 5x^2 + 4x \\
 x-3 \overline{) x^4 + 2x^3 - 11x} \\
 \underline{-(x^4 - 3x^3)} \\
 0 5x^3 - 11x \\
 \underline{-(5x^3 - 15x^2)} \\
 0 4x^2 - 11x \\
 \underline{-(4x^2 - 12x)} \\
 0 x
 \end{array}$$

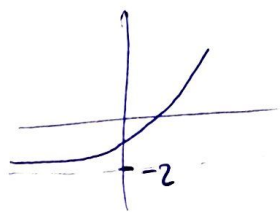
$$P(x) = (x-3) \cdot (x^3 + 5x^2 + 4x) + x$$

18. $2x^3 - 32x^2 < x^2 - 16$

$$2x^3 - 32x^2 - x^2 + 16 < 0$$

$$2x^3 - 33x^2 + 16 < 0$$

19. $g(x) = 2^x - 2$



domain = $(-\infty, \infty)$

range = $(-2, \infty)$

asymptote @ $y = -2$

20. a) $\ln(x+7) = 3$

$$e^{\ln(x+7)} = e^3$$

$$\underline{x+7 = e^3}$$

b) $\ln(x-9) = 5$

$$e^{\ln(x-9)} = e^5$$

$$\underline{x-9 = e^5}$$

21. $\log_2(40) - \log_2(10) = \log_2\left(\frac{40}{10}\right) = \log_2 4 = \log_2 2^2 = 2 \log_2 2 = \underline{\underline{2}}$

22. $\sin^{-1}\left(\frac{\sqrt{2}}{2}\right) = \frac{\pi}{4} \quad \tan \frac{\pi}{4} = 1$

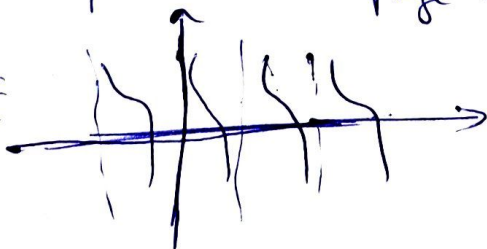
23. $f(x) = 3\sin x$



domain = $(-\infty, \infty)$

range = $[-3, 3]$

24. period = π



25.

